Riemann's Integration by Parts and the Explicit Formula for $J(x)$

Notes on Edwards' *Riemann's Zeta Function*

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Abstract

A comprehensive mathematical note on Bernhard Riemann's 1859 derivation of the explicit formula for the prime step-function $J(x)$. We detail the Mellin inversion of $\ln \zeta(s)$, contrast the divergent integration-by-parts choice with Riemann's convergent formulation, present a rigorous proof of the vanishing boundary terms along the vertical contour, and provide an expanded, step-by-step residue calculation over the non-trivial zeros ρ, trivial zeros, and principal pole of $\zeta(s)$.

1 The Mellin Inversion Setup

Riemann defined the prime counting step-function $J(x)$ as:

$$J(x) = \sum_{n=1}^{\infty} \frac{1}{n} \pi(x^{1/n}) = \pi(x) + \frac{1}{2}\pi(x^{1/2}) + \frac{1}{3}\pi(x^{1/3}) + \dots \tag{1}$$

where $\pi(x)$ is normalized at jump discontinuities via $\frac{1}{2}(\pi(x^+) + \pi(x^-))$. Taking the logarithm of the Euler product for $\zeta(s)$ for $\Re(s) > 1$ yields:

$$\ln \zeta(s) = \sum_{p \text{ prime}} \sum_{m=1}^{\infty} \frac{1}{mp^{ms}} = s \int_0^{\infty} J(x) x^{-s-1}\, dx. \tag{2}$$

Dividing by s, the expression $\frac{\ln \zeta(s)}{s}$ is recognized as the Mellin transform of $J(x)$. Applying Perron's formula / Mellin inversion along the vertical contour $\Re(s) = a > 1$:

$$J(x) = \frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} \frac{\ln \zeta(s)}{s} x^s\, ds. \tag{3}$$

2 The Integration by Parts Strategy

Directly attempting to evaluate integral (3) by choosing $u = \ln \zeta(s)$ and $dv = \frac{x^s}{s} ds$ requires integrating $\frac{x^s}{s}$, which generates exponential integrals $\mathrm{Ei}(s \ln x)$ inside the contour integral, leading to a divergent, intractable representation.

To circumvent this, Riemann flipped the Integration by Parts (IBP) direction on the truncated vertical segment $[a - iT, a + iT]$:

$$u(s) = \frac{\ln \zeta(s)}{s}, \qquad dv = x^s\, ds,$$

$$du = \left(\frac{\zeta'(s)}{s\zeta(s)} - \frac{\ln \zeta(s)}{s^2} \right) ds, \qquad v = \frac{x^s}{\ln x}.$$

Applying integration by parts yields:

$$\int_{a-iT}^{a+iT} \frac{\ln\zeta(s)}{s} x^s\, ds = \left[\frac{\ln\zeta(s)}{s}\frac{x^s}{\ln x}\right]_{a-iT}^{a+iT} - \frac{1}{\ln x}\int_{a-iT}^{a+iT} \frac{x^s}{\ln x}\, d\left(\frac{\ln\zeta(s)}{s}\right). \tag{4}$$

Expanding $d\left(\frac{\ln\zeta(s)}{s}\right) = \frac{\zeta'(s)}{s\zeta(s)}ds - \frac{\ln\zeta(s)}{s^2}ds$, the contour integral decomposes into:

$$J(x) = \lim_{T\to\infty}\frac{1}{2\pi i}\left[\frac{\ln\zeta(s)}{s}\frac{x^s}{\ln x}\right]_{a-iT}^{a+iT} - \frac{1}{2\pi i\ln x}\int_{a-i\infty}^{a+i\infty}\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\, ds + \frac{1}{2\pi i\ln x}\int_{a-i\infty}^{a+i\infty}\frac{\ln\zeta(s)}{s^2}x^s\, ds. \tag{5}$$

3 Vanishing of the Boundary Term

To evaluate the limit as $T\to\infty$, consider the integrated boundary increment:

$$I_B(T) = \left[\frac{\ln\zeta(s)}{s}\frac{x^s}{\ln x}\right]_{s=a-iT}^{s=a+iT}. \tag{6}$$

Lemma 1. For $a > 1$, $\lim_{T\to\infty} I_B(T) = 0$.

Proof. For $s = a + iT$ with $a > 1$:

$$|\ln\zeta(a+iT)| = \left|\sum_p\sum_{m=1}^{\infty}\frac{1}{mp^{m(a+iT)}}\right| \leq \sum_p\sum_{m=1}^{\infty}\frac{1}{mp^{ma}} = \ln\zeta(a) < \infty.$$

The numerator magnitude $|x^{a+iT}| = x^a$ is constant with respect to T. The denominator satisfies $|s| = \sqrt{a^2+T^2}$. Thus:

$$|I_B(T)| \leq \frac{2x^a}{\ln x}\cdot\frac{\ln\zeta(a)}{\sqrt{a^2+T^2}} \xrightarrow{T\to\infty} 0.$$

□

4 Contour Integration and Residue Calculus for $\frac{\zeta'(s)}{\zeta(s)}$

With the boundary term proven to vanish, the primary contribution to $J(x)$ stems from:

$$J_1(x) = -\frac{1}{2\pi i\ln x}\int_{a-i\infty}^{a+i\infty}\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\, ds. \tag{7}$$

To compute $J_1(x)$, we employ Hadamard's product expansion for the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. Taking the logarithmic derivative gives:

$$\frac{\zeta'(s)}{\zeta(s)} = -\frac{1}{s} - \frac{1}{s-1} + \frac{1}{2}\ln\pi - \frac{1}{2}\frac{\Gamma'(s/2+1)}{\Gamma(s/2+1)} + \sum_\rho\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right), \tag{8}$$

where the sum runs over all non-trivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$, ordered symmetrically by $|\gamma|$.

4.1 Evaluation via Contour Enclosure and Residues

We enclose the line of integration $\Re(s) = a > 1$ by a large rectangular contour C_R extending into the left half-plane ($\Re(s)\to-\infty$). By Cauchy's Residue Theorem:

$$\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\, ds = \sum\mathrm{Res}\left[\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\right]. \tag{9}$$

We compute the residues pole by pole:

4.1.1 1. The Pole at $s = 1$

Near $s = 1$, $\zeta(s)$ has a simple pole with residue 1, so $\zeta(s) = \frac{1}{s-1} + O(1)$ and $\frac{\zeta'(s)}{\zeta(s)} = -\frac{1}{s-1} + O(1)$.

$$\mathrm{Res}_{s=1}\left[\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\right] = \lim_{s\to 1}(s-1)\left(-\frac{1}{s-1}\right)\frac{x^s}{s} = -x.$$

Multiplying by the prefactor $-\frac{1}{\ln x}$ in $J_1(x)$ gives $\frac{x}{\ln x}$. Integrating this contribution with respect to $\ln x$ yields the fundamental term:

$$\mathrm{Li}(x) = \int_0^x \frac{dt}{\ln t}. \tag{10}$$

4.1.2 2. The Non-Trivial Zeros $s = \rho$

At each non-trivial zero $s = \rho$ of multiplicity 1, $\zeta(s)$ has a simple zero, so $\frac{\zeta'(s)}{\zeta(s)} = \frac{1}{s-\rho} + O(1)$.

$$\mathrm{Res}_{s=\rho}\left[\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\right] = \frac{x^\rho}{\rho}.$$

Including the prefactor $-\frac{1}{\ln x}$ and integrating gives $-\mathrm{Li}(x^\rho)$. Summing over all non-trivial zeros ρ yields:

$$-\sum_\rho \mathrm{Li}(x^\rho) = -\sum_{\Im(\rho)>0}\left(\mathrm{Li}(x^\rho) + \mathrm{Li}(x^{\bar\rho})\right). \tag{11}$$

4.1.3 3. The Trivial Zeros at $s = -2n$ ($n \in \mathbb{N}$)

The poles of $\Gamma(s/2+1)$ at $s = -2, -4, -6, \ldots$ correspond to the trivial zeros of $\zeta(s)$. At $s = -2n$:

$$\mathrm{Res}_{s=-2n}\left[\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s}\right] = \frac{x^{-2n}}{-2n}.$$

Multiplying by $-\frac{1}{\ln x}$ and summing over $n \geq 1$:

$$\sum_{n=1}^{\infty} \mathrm{Li}(x^{-2n}) = \int_x^\infty \frac{dt}{t(t^2-1)\ln t}. \tag{12}$$

4.1.4 4. The Pole at $s = 0$

At $s = 0$, $\frac{x^s}{s}$ has a simple pole with residue 1, while $\frac{\zeta'(0)}{\zeta(0)} = \ln(2\pi)$. Taking into account the second term in (5), $\frac{1}{2\pi i \ln x}\int \frac{\ln\zeta(s)}{s^2}x^s ds$, the contribution at $s = 0$ simplifies to the constant term:

$$-\ln 2. \tag{13}$$

5 The Complete Explicit Formula

Combining all residue contributions from Section 4 yields Bernhard Riemann's celebrated explicit formula for $J(x)$:

$$J(x) = \mathrm{Li}(x) - \sum_\rho \mathrm{Li}(x^\rho) - \ln 2 + \int_x^\infty \frac{dt}{t(t^2-1)\ln t}. \tag{14}$$

From $J(x)$, $\pi(x)$ is subsequently recovered by Möbius inversion:

$$\pi(x) = \sum_{n=1}^{\infty} \frac{\mu(n)}{n} J(x^{1/n}). \tag{15}$$